

Sample Project Documentation Summary

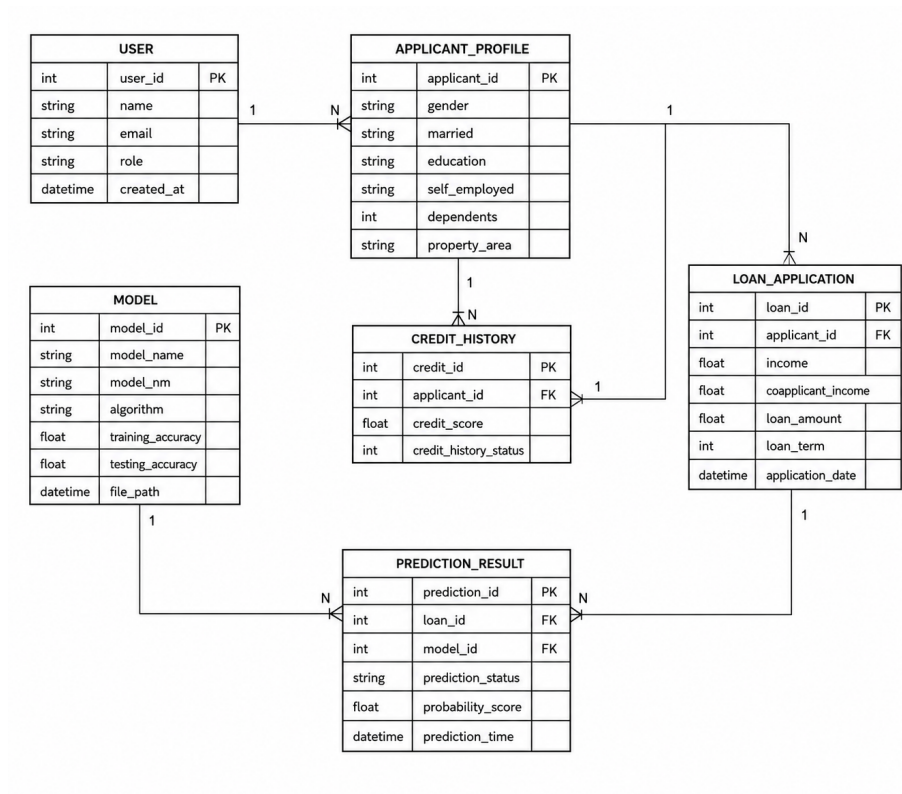
Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Sample Project Documentation Summary
Project Phase	Phase 7: Project Documentation
Submitted By	Team Leader: Palli Dharani Team Members: Yarabarla Manidweep Saikiran Kudipudi Sita Rama Raju Datla Thota Leela Sai Krishna
Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Abstract Overview

Smart Lender leverages a machine learning backend inside a Flask configuration to automate standard mortgage credit profiling. It processes demographic and financial parameters to predict eligibility status with minimal delay.

2. Database Schema Diagram (ERD)



3. Modeling Pipeline - SMOTE Class Balance

```
[ ] #creating a new x and y variables for the balanced set
    x_bal,y_bal = smote.fit_resample(x,y)

[ ] #printing the values of y before balancing the data and after
    print(y.value_counts())
    print(y_bal.value_counts())

[ ] names = x_bal.columns
```

Figure: Oversampling minority cases using SMOTE

4. Final Model Evaluation Matrix

Evaluation Case	Precision Rate	Recall Rate	F1 Score Metric
Approved (Class 1)	0.87	0.91	0.89
Rejected (Class 0)	0.83	0.78	0.80
Macro Average	0.85	0.85	0.85